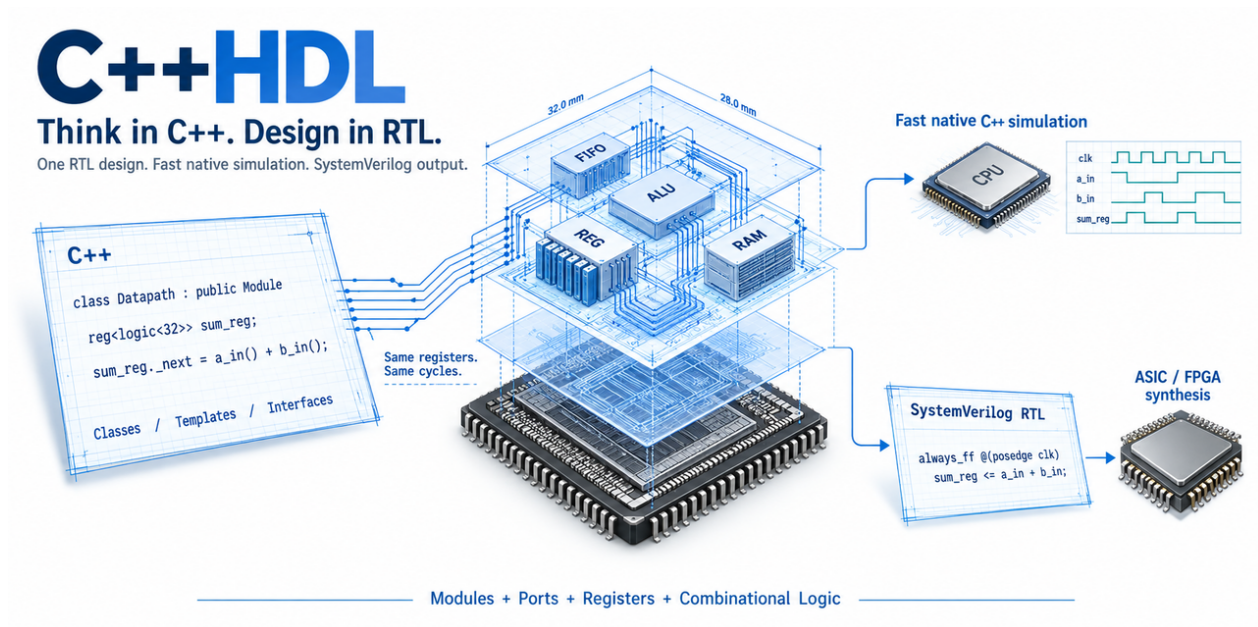
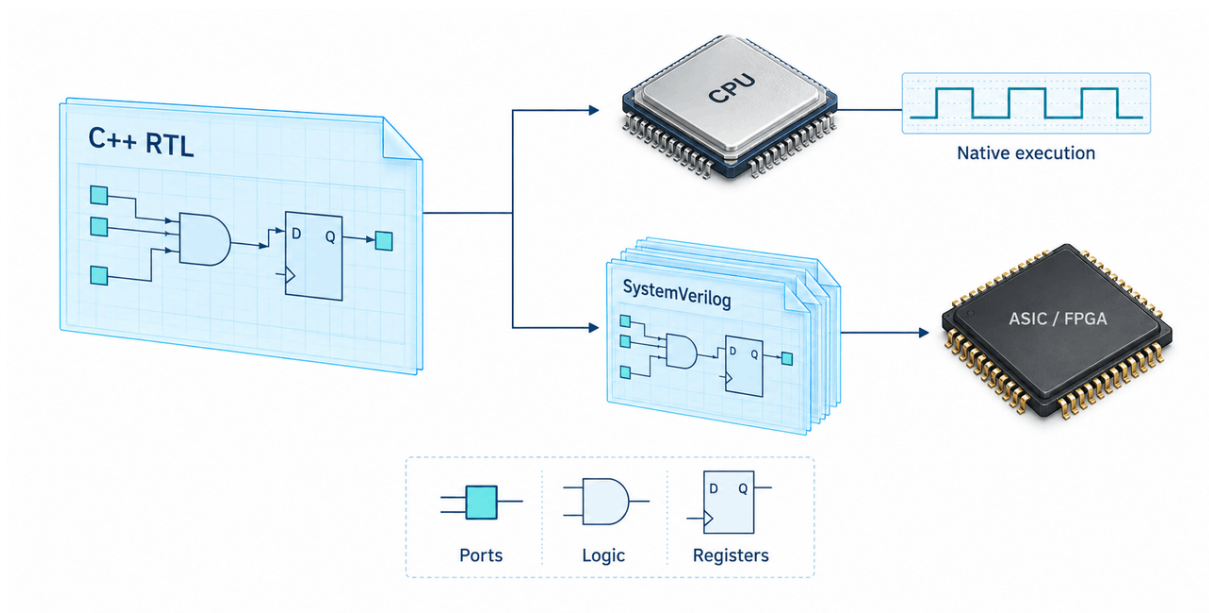


Designing RTL with C++HDL

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2026





1. Introduction

1.1 The design we are going to build

A sensor produces one-byte measurements. Its consumer sometimes pauses. We first use one clock for both, then implement a queue with separate write and read clocks.

We will implement this familiar RTL task in C++, then run the same description as a native executable and convert it to SystemVerilog. The four steps introduce the CppHDL constructs needed for each implementation:

- 1 Chapter 2: sensor -> threshold/capture stage -> consumer
- 2 Chapter 3: sensor -> capture stage -> memory queue -> consumer
- 3 Chapter 4: sensor -> source endpoint <-> queue endpoint <-> consumer endpoint
- 4 Chapter 5: write-clock endpoint -> asynchronous memory FIFO -> read-clock endpoint

Chapter 5 develops and tests a queue whose input and output use different clocks. We test this queue separately, then explain how to connect it to the earlier modules. We do not build a complete two-clock version of the earlier design.

CppHDL describes RTL in C++. A class represents a module, a member object can represent a child module, a function can represent combinational logic, and a `reg<T>` object stores a value that changes at a clock edge. The converter emits SystemVerilog. The same hardware description can also execute as a native C++ model.

The hardware concepts remain those of SystemVerilog RTL. What changes is how we express them and execute the model: sequential C++ calls must preserve concurrent hardware behavior. Much of this book explains that mapping.

This is **not high-level synthesis**. A C++ loop does not automatically become a multi-cycle accelerator, a method call does not consume a clock, and an ordinary variable does not become a pipeline register just because it appears inside a class. We explicitly choose where to place registers and which calculations happen between clock edges.

1.2 Seven reasons to use C++ for RTL

1. **Run the RTL directly as C++.** Compile the design with its testbench and use a C++ debugger, sanitizer, or profiler without an intermediate RTL conversion. Native execution is still simulation; measure its speed on your design rather than assuming a universal speedup.
2. **Generate RTL from C++ type parameters.** `Block<TYPE>` can use the selected type's fields, constants, and supported methods; CppHDL generates the corresponding RTL. The benefit is using C++ templates and specialization in the executable design, not merely parameterizing a width.
3. **Use the C++ ecosystem for verification and analysis.** Link the model into existing C++ test systems. Use C++ containers in the testbench to store and retrieve debug information as the simulation runs, and check it on each call that simulates an edge. Tools that inspect C++ can also analyze the design's classes and connections, check project coding rules, or draw diagrams.
4. **Give coding assistants a direct compile-and-test loop.** They can modify C++ RTL and run focused native tests without first generating SystemVerilog. This can shorten iteration; it does not remove the need for independent review.
5. **Share the implementation with software firmware.** Firmware and system-model teams can embed the C++ RTL directly instead of maintaining another handwritten peripheral model that may behave differently.
6. **Build large, complex multithreaded or cluster-based RTL simulations.** Use C++ threading and communication libraries to distribute model instances or independent test runs across CPU/GPU cores and machines. For connected models, the simulation framework must coordinate data exchange and simulation time.
7. **Prototype behavior first, then refine it into RTL.** Start with a non-synthesizable C++ sketch: use function calls as connections and pass objects, complete transactions, or memory buffers, as in transaction-level modeling (TLM). Simulate the system's behavior before defining individual wires and registers. Then replace the behavioral operations step by step with clocked, register-to-register logic, keeping the sketch as a reference for tests. This refinement is a design task, not automatic RTL conversion.

These are benefits of the C++ workflow, not claims that structs, interfaces, parameterization, or verification are new to RTL. Conversion supports a subset of C++. At the end of development, the generated SystemVerilog still needs acceptance verification in a timing/event-driven simulator and synthesis testing with the target synthesis tools.

1.3 The six-chapter plan

We will build and test a sample stage, add a memory queue, connect the modules through interfaces, and finally implement a two-clock FIFO. Each chapter introduces the CppHDL constructs needed for the next step.

Chapter	Example	CppHDL focus	Check
1. Introduction	The development workflow	Native execution and conversion	Build prerequisites
2. Capture a sample	A one-slot register stage	Ports, comb methods, work/strobe, and native VCD	Capture, hold, and drain
3. Absorb a burst	A stage feeding a memory queue	<code>memory<></code> , deferred writes, and calls to child modules	Ordering under backpressure
4. Connect reusable modules	Interface endpoints and a counter	Direction conventions, <code>assignIf</code> , and RTL class inheritance	Reuse the test with another C++ top-level type
5. Cross clocks	A memory-backed asynchronous FIFO	Named clock and reset methods, and a shared C++/RTL test	Ordering, reset, and coincident edges
6. Conclusion	Review the examples	What native C++ execution changes	Remaining RTL checks

Five revisions of that plan

The plan was revised to: (1) keep one realistic task, (2) state each example's transfer rules before its code, (3) demonstrate native execution immediately, (4) introduce child modules before interface inheritance, and (5) reuse checks across implementations.

1.4 Prerequisites and notation

This book assumes basic SystemVerilog RTL knowledge and familiarity with C++ classes, functions, and references. Explanations focus on CppHDL's syntax and execution rules; no previous CppHDL experience is required.

The API reference is `spec.md`; coding guidance is `best_practice.md`. The repository's examples and tests are useful companions, especially the existing two-clock FIFO and the CDC regression.

The complete source listings are labeled by filename and belong in one working directory. Later chapters include earlier headers; they do not replace them. Unlabeled fragments illustrate an expression or a method and are not additional standalone files. The chapter images are conceptual views, not complete port lists or substitutes for the code.

Get and build CppHDL

Get the source from github.com/mirekez/cpphdl. On Linux, with Git and Conda installed, create the supplied dependency environment and build the converter:

```
1 git clone https://github.com/mirekez/cpphdl.git
2 cd cpphdl
3 conda env create --prefix ../.conda --file requirements.yaml
4 conda activate ../.conda
5 cmake -S . -B build -DCMAKE_BUILD_TYPE=Release -G "Unix Makefiles"
6 cmake --build build --target cpphdl --parallel 4
7 ./build/cpphdl --help
```

The result is `build/cpphdl`, the C++-to-SystemVerilog converter. To simulate natively, include `cpphdl.h` in the model and compile it with your C++ testbench, adding `include/` to the compiler's include path. Run that executable directly. To generate RTL, pass the model source to `cpphdl`, choose an output directory with `--generated-dir`, and put compiler options such as `-I` and `-D` after `--`. The following chapters show both flows with complete commands.

The book's native commands use `g++` with C++17 support. The supplied Conda environment provides Verilator and Make; a VCD viewer is optional. See the README for platform-specific setup details.

Prepare the book examples

Keep the example files together so later chapters can include earlier headers. From the repository root, set the working paths:

```
1 # Run this setup from the CppHDL repository root.
2 export CPPHDL_SRC="$PWD"
3 export CPPHDL="$CPPHDL_SRC/build/cpphdl"
4 export BOOK="$CPPHDL_SRC/build/book-check"
5 mkdir -p "$BOOK"
```

This creates the working directory, not the source files shown below. Build each test as a separate executable: each defines its own `main()` and, for the C++ model, its own `_system_clock`. Do not link the tests into one executable.

Chapter	Source files needed in \$BOOK	Executable to run
2	SampleStage.h, sample_test.cpp	sample_test
3	Earlier header plus MemoryQueue.h, TelemetryBuffer.h, buffer_test.cpp	buffer_test
4	Earlier headers plus InterfaceTelemetry.h; reuse buffer_test.cpp	interface_test
5	AsyncSamples.h, async_test.cpp; independent of the earlier headers	async_native, then VAsyncSamples

There are three different build activities; a successful result from one does not imply success in the others:

Activity	Tool and input	Result
Native simulation	g++ compiles a test and its C++ RTL headers	An executable that drives and checks the C++ model
RTL conversion	cpphdl parses the design header	SystemVerilog modules and supporting packages
RTL checking/execution	Verilator reads the generated SystemVerilog	Lint diagnostics, or a separately built simulation executable

The first complete commands appear after the chapter 2 files. Converter options such as `--generated-dir` go before `--`; parsing options such as `-I` go after it. `--lint-only` checks generated RTL without running stimulus. Chapter 5 adds the Verilator build-and-run flow.

CppHDL emits struct definitions in packages and adds `Predef_pkg.sv` for its supporting declarations. The commands list these before the importing modules.

1.5 Three rules for every module and testbench

These three rules say where to connect ports, where to assign register values, and which methods the test calls. All later examples use them.

Rule 1: Connect ports during setup

You MUST connect all ports before the first work call. A module defines connections in `_assign()`; output bindings may also appear at member declarations, as in the listings. The parent connects its immediate children, and the testbench connects the top-level inputs during setup.

Use `assignIf` for connections between complete interfaces, not separate assignments for their fields. Each endpoint defines the fields it drives locally. Connections MUST follow the module hierarchy, one level at a time.

Call the top-level `_assign()` once before simulation. Interface setup may call an endpoint's `_assign()` more than once, so repeating it MUST only re-establish the same connections, not reset state or repeatedly allocate resources. You MUST NOT reconnect ports or call `_assign()` in the simulation loop: bindings represent fixed RTL wiring. Change the values supplying the inputs instead. Objects referenced by bindings MUST remain alive while those bindings are used.

Rule 2: Assign next register's values in work

For a member `reg<T> r`, **assign `r._next` in `_work(reset)` or a helper it calls.** Reading `r` still returns the current value. `r.strobe()` copies `r._next` into `r`. If `_next` has not changed since the last strobe, `r` keeps its value. You MUST NOT assign `r` directly in the RTL or connect another module's input to `r._next` instead of `r`.

Call `r.strobe()` only from the module's `_strobe()`. Similarly, assign a memory row with `storage[index] = value` during work and call `storage.apply()` during strobe. Comb functions calculate results from inputs and current register values. They MUST NOT write registers or memory, call strobe/apply, or connect ports. Call the required comb function directly; no preparation function is needed.

Assign each register to one clock edge and put its updates in that clock's work/strobe pair. Put synchronous reset assignments to `_next` in work's reset branch. Asynchronous reset uses the separate handlers in section 5.7. C++ construction does not automatically zero registers or memory.

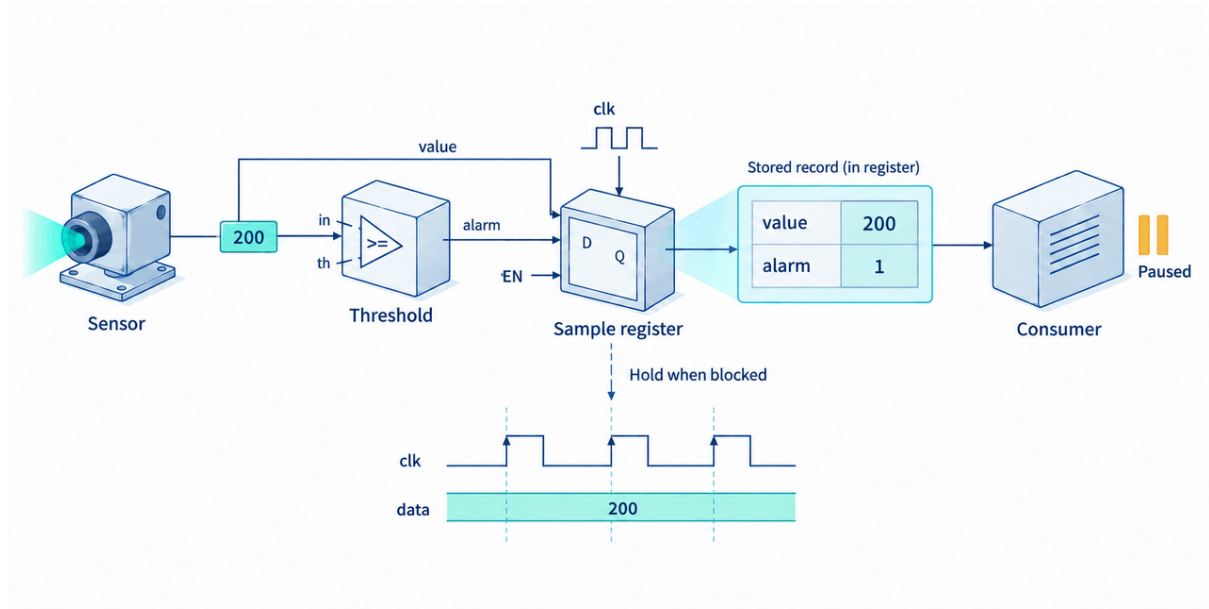
Rule 3: Call work and strobe methods through all hierarchy to make a clock tick

To simulate a clock edge in C++, your test MUST call the top module's `_work(reset)`, then `_strobe()`. The parent's `_work()` calls its children's work methods; the parent's `_strobe()` calls their strobe methods. The parent also passes reset to its children. You MUST NOT run work or strobe from `_assign()` or a comb function.

The order of children within work or within strobe does not matter. Keep the two phases separate: work calculates next values; strobe updates current values. When two clocks have an edge at the same simulation time, call both clocks' work methods before calling their strobe methods.

Increment `_system_clock` at the end of each simulation step.

Chapter 5 uses clock-specific method names and shows which additional calls are needed for asynchronous reset.



2. Capture a Sample

This example maps a one-slot RTL stage to CppHDL ports, combs, and registers, then runs a native test and writes VCD without generating SystemVerilog first.

2.1 The example's behavior

Suppose a sensor provides an eight-bit measurement. We want to flag values at or above a threshold and deliver the measurement and flag together.

The combinational decision is small:

```

1 alarm = measurement >= threshold
2 accept = enable AND input_valid AND space_available

```

Capture the measurement and its alarm flag together. `en_in` controls capture, not the clock signal.

2.2 Define when a sample transfers

The example uses rising-edge valid/ready handshakes. In CppHDL, parentheses read a port's current value:

- The stage accepts an input when `valid_in() && ready_out()` is true.

- The consumer accepts the output when `valid_out() && ready_in()` is true.

The stage can deliver its stored sample and capture the next sample on the same edge. Setting `en_in` to false blocks new inputs but still lets a stored sample leave.

The payload is one struct with two byte fields:

Field	Meaning
<code>value</code>	The measurement accepted on the input edge
<code>alarm</code>	Zero or one, calculated from the threshold on that same edge

A byte-sized flag keeps the C++ layout and later 16-bit encoding simple.

2.3 Build the module

Derive the class from `Module`; declare public signals with `_PORT(T)` and direction suffixes `_in` or `_out`. Use `reg<SampleWord>` for the payload and `reg<u1>` for valid. A comb result variable does not become a hardware register.

The listing uses these types:

C++ spelling	Meaning here
<code>u<8></code>	An unsigned eight-bit value
<code>u1</code>	A one-bit unsigned class type, suitable inside <code>reg<></code>
<code>uint8_t</code>	A native byte used for each field of the sample struct
<code>SampleWord</code>	The struct holding the sample; <code>__PACKED</code> requests packed C++ layout
<code>_PORT(SampleWord)</code>	A signal carrying a <code>SampleWord</code> value; it does not store that value between edges

Later, `logic<16>` provides a sixteen-bit value with bit-selection methods. For RTL values, use types with the intended width rather than a host-dependent type such as `size_t`. `reg<T>` requires a class type, so use `u1` rather than native `bool` as its template argument.

Declare a member variable for each comb result, followed by a method named `<variable_name>_func()`. For example, `SampleWord sample_comb;` is followed by `SampleWord& sample_comb_func()`. The method calculates the result on each direct call and returns a reference to that member variable. Assign every result field on every path.

Here is a reading key for the remaining syntax in the listing:

Construct	Purpose
<code>port = _ASSIGN(...)</code>	Install a connection that supplies a value when the port is read
<code>_ASSIGN_REG(...), _ASSIGN_COMB(...)</code>	Faster alternatives for assigning a register's value or a comb's result

You can use simple `_ASSIGN()` for any of these port connections if you are unsure. `_ASSIGN_REG(...)` and `_ASSIGN_COMB(...)` just make native simulation faster for register values and comb results, respectively.

The execution methods follow section 1.5. Here `_assign()` is empty because the stage's output bindings are at their declarations and its inputs are connected by the testbench.

The complete stage follows; section 2.4 then walks through its behavior.

File: SampleStage.h

```

1  #pragma once
2  #include <cpphdl.h>
3  using namespace cpphdl;
4
5  struct SampleWord
6  {
7      uint8_t value;
8      uint8_t alarm;
9  } __PACKED;
10
11 class SampleStage : public Module
12 {
13 public:
14     _PORT(bool) en_in;
15     _PORT(bool) valid_in;
16     _PORT(u<8>) value_in;
17     _PORT(u<8>) threshold_in;
18     _PORT(bool) ready_in;
19     _PORT(bool) ready_out = _ASSIGN_COMB(ready_comb_func());
20     _PORT(bool) valid_out = _ASSIGN((bool)valid_reg);
21     _PORT(SampleWord) sample_out = _ASSIGN_REG(sample_reg);
22
23 private:
24     reg<SampleWord> sample_reg;
25     reg<u1> valid_reg;
26
27     SampleWord sample_comb;
28     SampleWord& sample_comb_func()
29     {
30         sample_comb.value = (uint8_t)value_in();
31         sample_comb.alarm = value_in() >= threshold_in() ? 1 : 0;
32         return sample_comb;
33     }

```

```

34
35     bool ready_comb;
36     bool& ready_comb_func()
37     {
38         ready_comb = en_in() && (!(bool)valid_reg || ready_in());
39         return ready_comb;
40     }
41
42 public:
43     void _assign() {}
44
45     void _work(bool reset)
46     {
47         if (reset) {
48             sample_reg._next = {};
49             valid_reg._next = 0;
50         }
51         else if (!(bool)valid_reg || ready_in()) {
52             valid_reg._next = en_in() && valid_in();
53             if (en_in() && valid_in()) {
54                 sample_reg._next = sample_comb_func();
55             }
56         }
57     }
58
59     void _strobe()
60     {
61         sample_reg.strobe();
62         valid_reg.strobe();
63     }
64 };

```

2.4 C++ calls and register updates

The two combs answer separate questions:

- `sample_comb_func()` forms a complete payload from the current input.
- `ready_comb_func()` decides whether the stage can accept that payload on the next edge.

`_work` uses `sample_comb_func()` when capturing a sample; the ready output uses `ready_comb_func()` through its binding.

You can use `_ASSIGN()` for all these port connections. If you are unsure whether an expression refers to register storage or a persistent comb result, use `_ASSIGN()`. `_ASSIGN_REG()` and `_ASSIGN_COMB()` are optional alternatives that make native simulation faster for register values and comb results.

Binding	Appropriate use
<code>_ASSIGN(expression)</code>	General choice, including register values, comb results, casts, and Boolean expressions
<code>_ASSIGN_REG(object)</code>	Faster binding to persistent register or input storage
<code>_ASSIGN_COMB(function())</code>	Faster binding to a comb result returned by reference

The last two bind an **lvalue**, an existing object rather than a copied value. A temporary such as `u<8>(128)` or a function result returned by value does not meet their lifetime requirement; use `_ASSIGN()` for those expressions.

The reset branch assigns zero to `sample_reg._next` and `valid_reg._next`. When the stage is full and the consumer is not ready, neither non-reset assignment runs. Both registers keep their values according to Rule 2.

The conditions produce these results:

Situation before the edge	Action at the edge
Reset asserted	Clear the payload and valid bit
Full and downstream not ready	Hold payload and valid, regardless of enable
Slot available, enabled, input valid	Capture the new payload and set valid
Slot available, but no enabled valid input	Clear valid; the old payload bits need not change

With no clock options, the converter supplies `clk` and an active-high `reset`. `_work(true)` models synchronous reset; `_strobe()` applies it to both registers. No explicit C++ clock port is needed in this single-clock mode.

After conversion, expect a struct package, module ports, combinational logic, and an `always_ff @(posedge clk)` process. The converter may express the body through generated tasks and temporary variables for the next register values.

2.5 Execute one edge correctly

Native simulation uses a global counter to decide when cached port values must be recalculated:

```
1 long _system_clock = -1;
```

Define it once per executable. The converter does not emit this counter into RTL. The single-clock test uses Rule 3 in this order:

```

1 set testbench stimulus values, if needed
2 advance _system_clock
3 observe transfers that will happen on this edge
4 call _work(reset)
5 call _strobe()
6 advance _system_clock
7 observe the newly committed outputs

```

In the test, the input variables are members, the constructor connects them, and `tick(true)` performs reset. Later `tick()` calls default to normal operation. This test reads outputs immediately after strobe, so it increments the counter before those reads. It also increments the counter at the next `tick()` to discard results cached by those checks before evaluating new stimulus. The checks cover:

Edge	Input situation	State after the edge
Reset	No valid sample	Empty
Capture	Value 200, threshold 128, downstream blocked	Holds 200 with alarm 1
Stall	Next input is 17, downstream still blocked	Still holds 200
Drain	Input disabled, downstream ready	Empty
Capture	Enabled again, input 17	Holds 17 with alarm 0
Drain	No new valid input	Empty

2.6 Write the first waveform

CppHDL's `VcdFile` reads values from addresses supplied by the testbench. We copy the three outputs into ordinary testbench variables and give their addresses to `VcdFile`. We do not pass port objects, comb results, or whole `reg<T>` objects. A `reg<T>` object also stores the next value, so its C++ object size is larger than the hardware value we want to display.

The test copies the outputs after calling `_strobe()` and incrementing `_system_clock`. The variables remain alive until the `VcdFile` destructor closes the file. The test writes one set of output values every 10 ns; it does not record intermediate combinational changes.

Each signal entry is `{name, bit_width, address_of_storage}`. The `shown_*` fields hold these copies in the testbench; they are not additional RTL registers. The minimal dump includes the stored value, alarm, and valid bit, but does not include the input signals or a clock lane. The chapter illustration shows the timing concept; it is not a screenshot of this VCD.

File: `sample_test.cpp`

```

1  #include "SampleStage.h"
2  #include <cpphdl_vcd.h>
3  #include <cassert>
4  #include <cstdio>
5
6  long _system_clock = -1;
7
8  struct SampleTest
9  {
10     SampleStage dut;
11     bool enable = true;
12     bool valid = false;
13     bool ready = false;
14     u<8> value = 0;
15     u<8> threshold = 128;
16     uint8_t shown_value = 0;
17     uint8_t shown_alarm = 0;
18     uint8_t shown_valid = 0;
19     unsigned time_ns = 0;
20     VcdFile trace;
21
22     SampleTest()
23     {
24         dut.en_in = _ASSIGN(enable);
25         dut.valid_in = _ASSIGN(valid);
26         dut.ready_in = _ASSIGN(ready);
27         dut.value_in = _ASSIGN_REG(value);
28         dut.threshold_in = _ASSIGN_REG(threshold);
29         dut._assign();
30         trace.signals = {
31             {"value", 8, &shown_value},
32             {"alarm", 8, &shown_alarm},
33             {"valid", 1, &shown_valid}
34         };
35         trace.create("samples.vcd");
36     }
37
38     void tick(bool reset = false)
39     {
40         ++_system_clock;
41         dut._work(reset);
42         dut._strobe();
43         ++_system_clock;
44         shown_value = dut.sample_out().value;
45         shown_alarm = dut.sample_out().alarm;
46         shown_valid = dut.valid_out();
47         trace.sample(time_ns);
48         time_ns += 10;
49     }
50 };
51
52 int main()
53 {

```

```

54 SampleTest test;
55 test.tick(true);
56 assert(!test.dut.valid_out());
57
58 test.valid = true;
59 test.value = 200;
60 test.tick();
61 assert(test.dut.valid_out());
62 assert(test.dut.sample_out().value == 200);
63 assert(test.dut.sample_out().alarm == 1);
64
65 test.value = 17;
66 test.tick();
67 assert(!test.dut.ready_out());
68 assert(test.dut.sample_out().value == 200);
69
70 test.enable = false;
71 test.ready = true;
72 test.tick();
73 assert(!test.dut.valid_out());
74
75 test.enable = true;
76 test.tick();
77 assert(test.dut.sample_out().value == 17);
78 assert(test.dut.sample_out().alarm == 0);
79
80 test.valid = false;
81 test.tick();
82 assert(!test.dut.valid_out());
83 std::puts("PASS: capture, hold, drain, enable; wrote samples.vcd");
84 }

```

2.7 Build and test the sample stage

Compile and run `sample_test.cpp` to check capture, hold, drain, and enable behavior and produce a VCD file. Then convert `SampleStage.h` to SystemVerilog and use Verilator to check the generated RTL without running it.

```

1 g++ -std=c++17 -O2 -I"$CPPHDL_SRC/include" \
2   "$BOOK/sample_test.cpp" -o "$BOOK/sample_test"
3 (cd "$BOOK" && ./sample_test)
4
5 "$CPPHDL" "$BOOK/SampleStage.h" --generated-dir="$BOOK/sv_stage" -- \
6   -I"$CPPHDL_SRC/include"
7
8 verilator --lint-only --top-module SampleStage \
9   "$BOOK/sv_stage/Predef_pkg.sv" \
10  "$BOOK/sv_stage/SampleWord_pkg.sv" \
11  "$BOOK/sv_stage/SampleStage.sv"

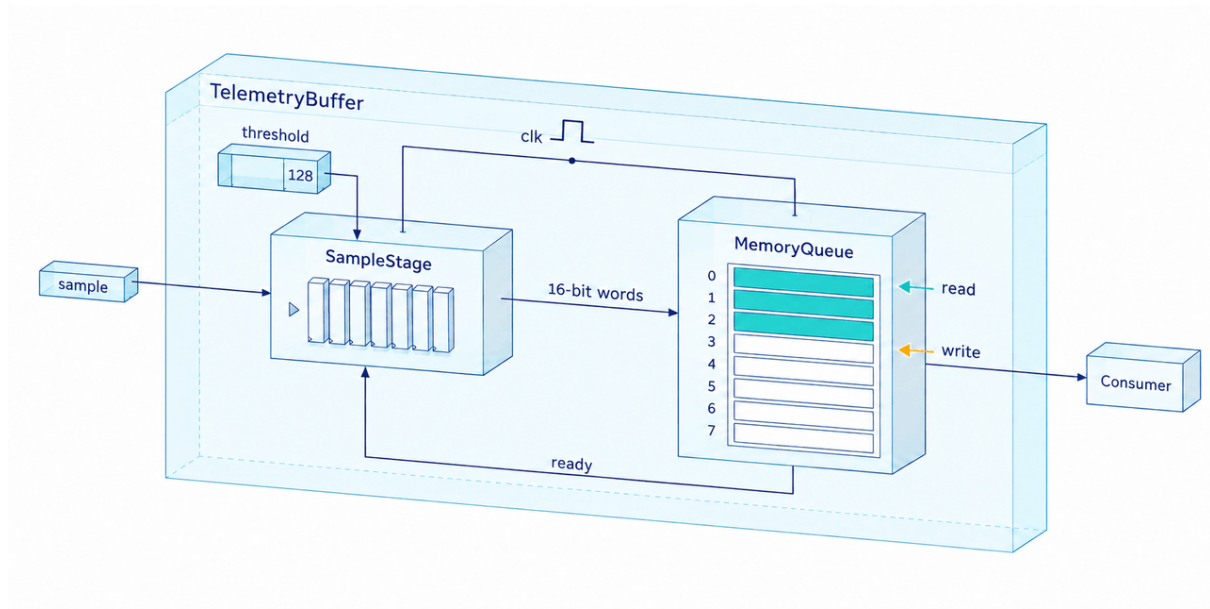
```

Keep C++ assertions enabled: do not define `NDEBUG`. In `$BOOK/samples.vcd`, the output holds 200 at 10 and 20 ns, becomes invalid at 30 ns, and contains 17 at 40 ns. The test drives the inputs, but this VCD traces only outputs.

The native executable should print `PASS: capture, hold, drain, enable; wrote samples.vcd`. The conversion and lint commands check that the design can also be represented as RTL. They do not yet run this test against that RTL.

Check your CppHDL understanding: Why does changing a bound input require incrementing `_system_clock`? Why does this test copy outputs into `shown_*` variables instead of dumping the whole `reg<T>` object?

The next chapter adds a queue so the input can accept several more samples while the consumer is paused.



3. Absorb Bursts with Memory and Child Modules

This chapter adds `memory<>` and child modules. The main CppHDL concern is preserving simultaneous register and memory updates despite sequential C++ calls.

3.1 The queue behavior used here

Add an eight-word queue after the stage. Its behavior is:

Question	Choice in this chapter
How much queue storage?	Eight words, in addition to the stage's one slot
What is a word?	Sixteen bits: value in bits 7:0, alarm byte in bits 15:8
When is a word visible?	Combinationally from the current head while nonempty
What happens when full?	Input ready is low, even if a read occurs on this edge

Here input ready depends on the stored count, not on output ready. A full queue therefore cannot accept a replacement on the edge that removes a word. The example uses show-ahead reads; `memory<>` does not automatically adapt that behavior to synchronous-read RAM.

3.2 Use `memory<>` and width-dependent C++ types

We need to store each measurement and its alarm byte together. Declare a memory with two bytes per row and use `DEPTH` to select the number of rows:

```
1 memory<u8, 2, DEPTH> storage;
```

That means `DEPTH` rows, each containing **two byte elements**. The second argument is not a bit count. This differs from `array<COUNT, TYPE, PACKED>`, whose first argument is an element count.

Here `u8` is CppHDL's byte class, and `MemoryQueue<8>` selects eight rows. This example demonstrates a numeric template parameter, not the type-based specialization discussed in the introduction.

With `memory<>`, assigning a row adds it to a list of pending writes. `storage.apply()` copies those writes into the stored rows and clears the list. A normal read still returns the stored row, not the pending write, as required by Rule 2.

The usual pointer and count widths become C++ type expressions: `u<clog2(DEPTH)>` and `u<clog2(DEPTH) + 1>`. `static_assert` restricts this implementation to supported power-of-two depths.

Use `uint32_t` when indexing RTL storage. A host `size_t` may be 64 bits; unnecessarily wide runtime RTL indices can cause synthesis-tool problems. Template size parameters can remain `size_t`.

3.3 Implement the queue

When the queue is empty, the module returns zero on its data output and false on its valid output. The data comb reads memory only when the count is nonzero, so it does not read an uninitialized row after reset.

This queue's reset clears pointers and count without reading or clearing the RAM.

File: `MemoryQueue.h`

```
1  #pragma once
2  #include <cpphdl.h>
3  using namespace cpphdl;
4
5  template<size_t DEPTH>
6  class MemoryQueue : public Module
7  {
8  public:
9      _PORT(bool) valid_in;
10     _PORT(logic<16>) data_in;
```

```

11  _PORT(bool) ready_in;
12  _PORT(bool) ready_out = _ASSIGN((uint32_t)count_reg < DEPTH);
13  _PORT(bool) valid_out = _ASSIGN((uint32_t)count_reg != 0);
14  _PORT(logic<16>) data_out = _ASSIGN_COMB(data_comb_func());
15
16  private:
17      static_assert(DEPTH >= 2 && (DEPTH & (DEPTH - 1)) == 0,
18          "DEPTH must be a power of two, at least two");
19      static constexpr size_t ADDR_BITS = clog2(DEPTH);
20      memory<u8, 2, DEPTH> storage;
21      reg<u<ADDR_BITS>> write_reg;
22      reg<u<ADDR_BITS>> read_reg;
23      reg<u<ADDR_BITS + 1>> count_reg;
24
25      logic<16> data_comb;
26      logic<16>& data_comb_func()
27      {
28          data_comb = 0;
29          if ((uint32_t)count_reg != 0) {
30              data_comb = (logic<16>)storage[(uint32_t)read_reg];
31          }
32          return data_comb;
33      }
34
35  public:
36      void _assign() {}
37
38      void _work(bool reset)
39      {
40          bool push;
41          bool pop;
42          if (reset) {
43              write_reg._next = 0;
44              read_reg._next = 0;
45              count_reg._next = 0;
46          }
47          else {
48              push = valid_in() && ready_out();
49              pop = valid_out() && ready_in();
50              if (push) {
51                  storage[(uint32_t)write_reg] = data_in();
52                  write_reg._next = write_reg + 1;
53              }
54              if (pop) {
55                  read_reg._next = read_reg + 1;
56              }
57              if (push && !pop) {
58                  count_reg._next = count_reg + 1;
59              }
60              else if (pop && !push) {
61                  count_reg._next = count_reg - 1;
62              }
63          }
64      }

```

```

65
66     void _strobe()
67     {
68         storage.apply();
69         write_reg.strobe();
70         read_reg.strobe();
71         count_reg.strobe();
72     }
73 };

```

The queue uses the same implicit `clk` and synchronous reset as the sample stage. `_assign()` is empty because the queue has no children or interface bundles; its output bindings live in the port declarations.

3.4 Add the stage and queue as members of a parent

The parent sets the threshold, converts the measurement and alarm flag into a 16-bit word, and connects the stage to the queue. We can still test either child module separately.

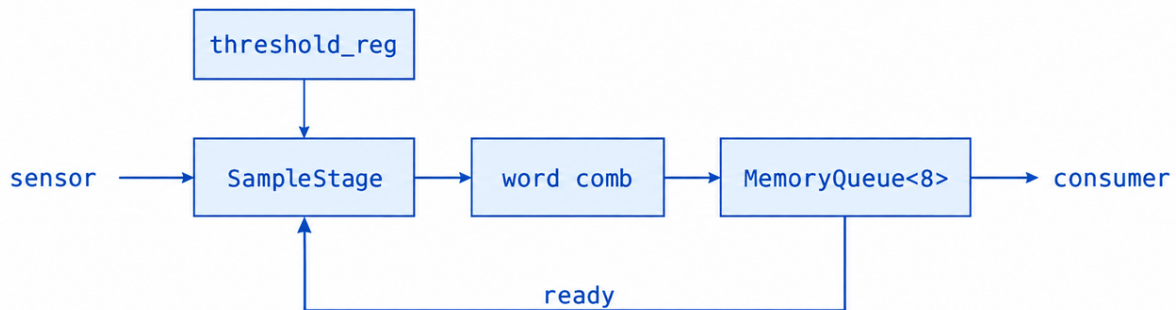


Figure 1: SampleStage receives the threshold and sends encoded samples to MemoryQueue; ready returns from the queue to the stage.

The parent uses three kinds of connections:

1. A child input can read a **parent input port**, such as the measurement.
2. A child input can read a **parent register**, such as the threshold.
3. A child input can read a **parent comb**, such as the encoded payload.

Use `word_comb.bits(high, low)` for inclusive bit selection, not a `reinterpret_cast` of the C++ struct. The encoding is value in the low byte and alarm in the high byte. The threshold register holds 128 after reset.

value_in() and the child port getters return references, so _ASSIGN_COMB can bind them too. Its argument must be an lvalue as described in section 2.4.

File: TelemetryBuffer.h

```
1  #pragma once
2  #include "SampleStage.h"
3  #include "MemoryQueue.h"
4
5  class TelemetryBuffer : public Module
6  {
7      SampleStage stage;
8      MemoryQueue<8> queue;
9
10     public:
11         _PORT(bool) en_in;
12         _PORT(bool) valid_in;
13         _PORT(u<8>) value_in;
14         _PORT(bool) ready_in;
15         _PORT(bool) ready_out = _ASSIGN(stage.ready_out());
16         _PORT(bool) valid_out = _ASSIGN(queue.valid_out());
17         _PORT(logic<16>) data_out = _ASSIGN_COMB(queue.data_out());
18
19     private:
20         reg<u<8>> threshold_reg;
21
22         logic<16> word_comb;
23         logic<16>& word_comb_func()
24         {
25             word_comb = 0;
26             word_comb.bits(7, 0) = stage.sample_out().value;
27             word_comb.bits(15, 8) = stage.sample_out().alarm;
28             return word_comb;
29         }
30
31     public:
32         void _assign()
33         {
34             stage.en_in = _ASSIGN(en_in());
35             stage.valid_in = _ASSIGN(valid_in());
36             stage.value_in = _ASSIGN_COMB(value_in());
37             stage.threshold_in = _ASSIGN_REG(threshold_reg);
38             stage.ready_in = _ASSIGN(queue.ready_out());
39             queue.valid_in = _ASSIGN(stage.valid_out());
40             queue.data_in = _ASSIGN_COMB(word_comb_func());
41             queue.ready_in = _ASSIGN(ready_in());
42             stage._assign();
43             queue._assign();
44         }
45
46         void _work(bool reset)
47         {
48             if (reset) {
```

```

49         threshold_reg._next = 128;
50     }
51     stage._work(reset);
52     queue._work(reset);
53 }
54
55 void _strobe()
56 {
57     threshold_reg.strobe();
58     stage._strobe();
59     queue._strobe();
60 }
61 };

```

3.5 Understand hierarchy and latency

TelemetryBuffer applies the three rules to a hierarchy: `_assign()` connects the children, `_work()` calls their work methods, and `_strobe()` calls their strobes. At E0, the stage captures the first sample while the queue still sees the stage's previous `valid_out == false`.

Starting empty, with the consumer ready, the first sample moves as follows:

Edge	Transfers at this edge	State visible after strobe
E0	Sensor sample enters the stage	Stage valid, queue empty
E1	Stage sample enters the queue	Queue valid; stage can hold the next sample
E2	Consumer removes the oldest queued sample	Following queued data, if any, becomes the head

The parent adds no register stage. Total capacity remains nine samples.

3.6 Link the model into a C++ test

The test directly instantiates the C++ RTL class and uses `std::deque<uint16_t>` as its scoreboard. Only RTL headers go to the converter; the host test can use ordinary C++ libraries without synthesis restrictions.

The test reads the output for its scoreboard before calling `_work()` and `_strobe()`, as in Rule 3.

The test sends 128 samples. It first holds the consumer's ready input low to fill the stage and queue, then pauses the consumer periodically. There are enough transfers for each queue pointer to wrap several times.

Leave `USE_INTERFACES` undefined for this chapter: `Design` then aliases `TelemetryBuffer`, and the later `InterfaceTelemetry.h` is not required. Chapter 4 changes that alias while reusing the test.

File: `buffer_test.cpp`

```
1  #ifndef USE_INTERFACES
2  #include "InterfaceTelemetry.h"
3  using Design = InterfaceTelemetry;
4  #else
5  #include "TelemetryBuffer.h"
6  using Design = TelemetryBuffer;
7  #endif
8  #include <cassert>
9  #include <cstdio>
10 #include <deque>
11
12 long _system_clock = -1;
13
14 int main()
15 {
16     Design dut;
17     bool enable = true;
18     bool valid = false;
19     bool ready = false;
20     u<8> value = 0;
21     std::deque<uint16_t> expected;
22     unsigned sent = 0;
23     unsigned received = 0;
24     bool stalled = false;
25
26     dut.en_in = _ASSIGN(enable);
27     dut.valid_in = _ASSIGN(valid);
28     dut.value_in = _ASSIGN_REG(value);
29     dut.ready_in = _ASSIGN(ready);
30     dut._assign();
31     ++_system_clock;
32     dut._work(true);
33     dut._strobe();
34     ++_system_clock;
35
36     for (unsigned cycle = 0; cycle < 2000; ++cycle) {
37         valid = sent < 128;
38         value = (uint8_t)(sent * 13u);
39         ready = cycle >= 30 && cycle % 7 != 0;
40         ++_system_clock;
41
42         const bool push = valid && dut.ready_out();
43         const bool pop = ready && dut.valid_out();
44         if (valid && !dut.ready_out()) stalled = true;
45         if (pop) {
46             assert(!expected.empty());
47             assert((uint16_t)dut.data_out() == expected.front());
```

```

48         expected.pop_front();
49         ++received;
50     }
51     if (push) {
52         const uint16_t byte = (uint8_t)value;
53         expected.push_back(byte | ((byte >= 128 ? 1u : 0u) << 8));
54         ++sent;
55     }
56     dut._work(false);
57     dut._strobe();
58     ++_system_clock;
59
60     if (received == 128) {
61         assert(sent == 128 && expected.empty() && stalled);
62 #ifdef USE_INTERFACES
63         assert((uint32_t)dut.sent_out() == received);
64 #endif
65         std::puts("PASS: 128 ordered samples under backpressure");
66         return 0;
67     }
68 }
69 std::fputs("FAIL: buffer did not drain\n", stderr);
70 return 1;
71 }

```

3.7 Test the buffer and inspect its generated RTL

Run `buffer_test.cpp` to check that the stage and queue deliver samples in order while the consumer pauses. Then convert `TelemetryBuffer.h` and lint the generated modules. Inspect the RTL to confirm that the stage, queue, and memory writes appear as intended.

```

1 g++ -std=c++17 -O2 -I"$CPPHDL_SRC/include" \
2     "$BOOK/buffer_test.cpp" -o "$BOOK/buffer_test"
3 "$BOOK/buffer_test"
4
5 "$CPPHDL" "$BOOK/TelemetryBuffer.h" --generated-dir="$BOOK/sv_buffer" -- \
6     -I"$CPPHDL_SRC/include" -I"$BOOK"
7
8 verilator --lint-only --top-module TelemetryBuffer \
9     "$BOOK/sv_buffer/Predef_pkg.sv" \
10    "$BOOK/sv_buffer/SampleWord_pkg.sv" \
11    "$BOOK/sv_buffer/SampleStage.sv" \
12    "$BOOK/sv_buffer/MemoryQueue.sv" \
13    "$BOOK/sv_buffer/TelemetryBuffer.sv"

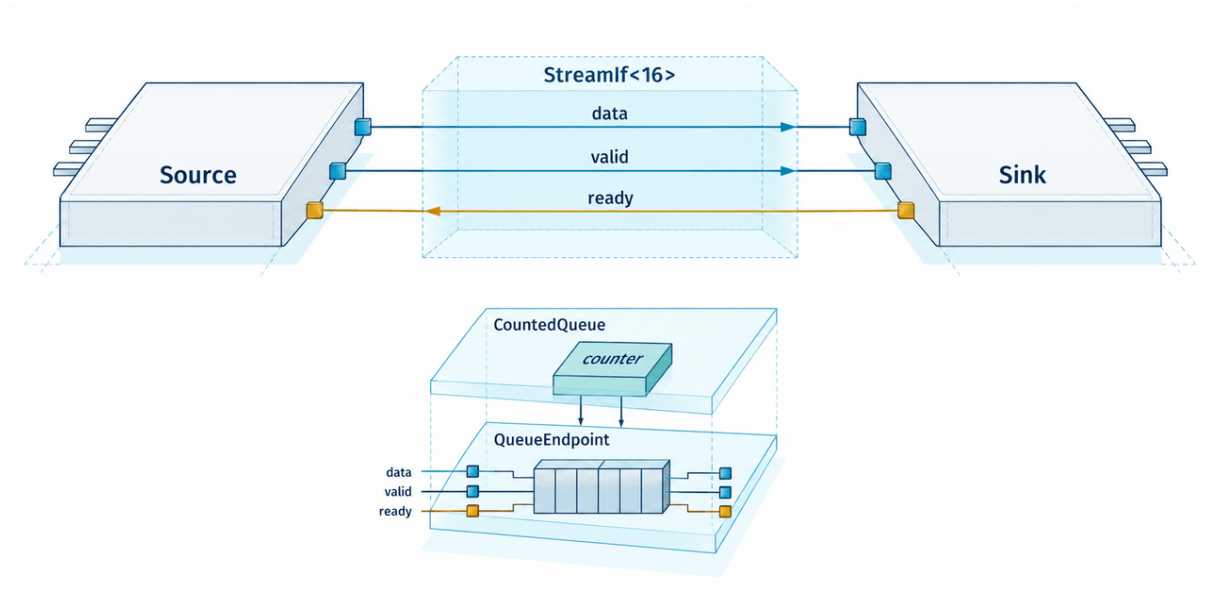
```

In the generated RTL, verify that the `stage` and `queue` members became child instances and that deferred memory writes became clocked assignments. `memory<>` is not a promise of a particular FPGA RAM or ASIC SRAM implementation.

The expected native result is `PASS: 128 ordered samples under backpressure`. The test checks ordering and observes backpressure. It does not assert exact E0/E1/E2 latency or exhaustively test full/empty transitions.

Check your CppHDL understanding: What would change if the parent called `stage._strobe()` before `queue._work()`? How would you detect that scheduling bug even if output ordering remained correct?

The hardware structure is now reusable. The connections are still expressed as individual scalar ports. The next chapter groups those ports into interfaces so developers can connect and extend modules without handling every wire separately.



4. Connect Reusable Interface Endpoints

This chapter replaces scalar connections with CppHDL interfaces and uses C++ inheritance to extend a synthesizable endpoint. The native test stays the same.

4.1 Declare a CppHDL interface

Derive `StreamIf<WIDTH>` from `Interface` and declare its signals with `_PORT`. It carries the same streaming protocol as before:

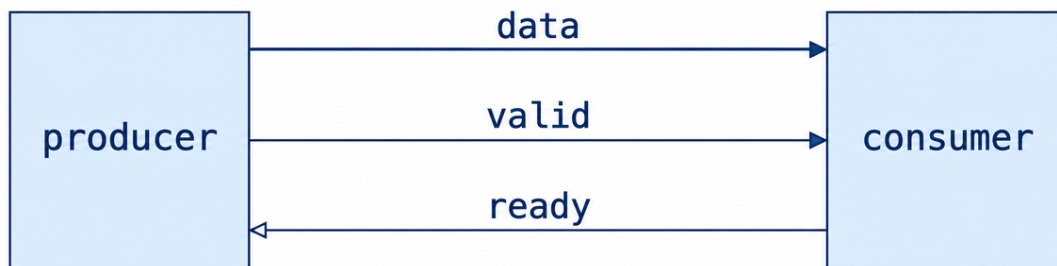


Figure 2: Data and valid travel from producer to consumer; ready travels back to the producer.

CppHDL determines interface directions from the member name's `_in` or `_out` suffix instead of a SystemVerilog modport. Both endpoints use the same C++ type.

4.2 Learn the two direction conventions

Declare `valid_in`, `data_in`, and `ready_out` in the interface type.

A member named `sink_in` keeps each field's declared direction. A member named `source_out` reverses every field's direction. In the table, direction is relative to the module containing the interface:

C++ access	Direction	Driven by
<code>sink_in.valid_in</code>	Input	Producer
<code>sink_in.data_in</code>	Input	Producer
<code>sink_in.ready_out</code>	Output	This module
<code>source_out.valid_in</code>	Output	This module
<code>source_out.data_in</code>	Output	This module
<code>source_out.ready_out</code>	Input	Consumer

The C++ field names do not change. To determine the direction of a generated SystemVerilog port, read both the containing interface member's name and the field's name. For example, the `_in` in `data_in` does not by itself mean that the containing module receives that signal.

Here **flattening** means emitting an individual module signal for each interface field. For example, the source's C++ `source_out.data_in` becomes an RTL output named `source_out__data_out`, while `source_out.ready_out` becomes an RTL input named `source_out__ready_in`.

The important CppHDL detail is that the source binds `source_out.data_in` despite its `_in` leaf suffix. Interface reuse itself is not a new RTL capability.

4.3 Divide work between components and subclasses

We will keep the existing capture and queue behavior:

- **SampleSource** **contains** the chapter 2 stage and exposes a stream output.
- **QueueEndpoint** **contains** the chapter 3 memory queue and exposes two stream interfaces.
- **CountedQueue** **inherits** from **QueueEndpoint** and counts the words the consumer accepts.
- **StreamMonitor** adapts the final interface to the scalar ports our testbench already knows.

Here C++ inheritance extends a synthesizable module. `CountedQueue::_work()` calls `QueueEndpoint::_work(reset)` and calculates the counter's next value. Its `_strobe()` calls `QueueEndpoint::_strobe()` and `sent_reg.strobe()`. These explicit base calls are necessary because the derived methods hide the base methods; C++ does not call both automatically.

The derived counter is synthesized hardware. Put it in the host test instead if it is needed only for simulation analysis.

Two small differences from chapter 3 are intentional. The top binds a constant threshold of 128 instead of storing 128 in a register that never changes; the alarm calculation after reset is unchanged. Also, `StreamMonitor` is only a port adapter despite its name. The host scoreboard, not that adapter, checks correctness.

4.4 Define local signals, then connect interfaces

`QueueEndpoint::_assign()` binds `sink_in.ready_out` to `queue.ready_out()` and its output data/valid fields to the queue's corresponding outputs.

A parent connects the two complete interfaces using:

```
1 assignIf(producer, buffer, producer.source_out, buffer.sink_in);
```

This is the interface form of Rule 1. `assignIf` connects data/valid toward the buffer and ready toward the producer, and calls their `_assign()` methods.

`CountedQueue` inherits `_assign()` and the `queue` member from `QueueEndpoint`.

File: InterfaceTelemetry.h

```
1  #pragma once
2  #include "SampleStage.h"
3  #include "MemoryQueue.h"
4
5  template<size_t WIDTH>
6  struct StreamIf : public Interface
7  {
8      _PORT(bool) valid_in;
9      _PORT(logic<WIDTH>) data_in;
10     _PORT(bool) ready_out;
11 };
12
13 class SampleSource : public Module
14 {
15     SampleStage stage;
16
17 public:
18     StreamIf<16> source_out;
19     _PORT(bool) en_in;
20     _PORT(bool) valid_in;
21     _PORT(u<8>) value_in;
22     _PORT(u<8>) threshold_in;
23     _PORT(bool) ready_out = _ASSIGN(stage.ready_out());
24
25 private:
```

```

26     logic<16> word_comb;
27     logic<16>& word_comb_func()
28     {
29         word_comb = 0;
30         word_comb.bits(7, 0) = stage.sample_out().value;
31         word_comb.bits(15, 8) = stage.sample_out().alarm;
32         return word_comb;
33     }
34
35 public:
36     void _assign()
37     {
38         stage.en_in = _ASSIGN(en_in());
39         stage.valid_in = _ASSIGN(valid_in());
40         stage.value_in = _ASSIGN_COMB(value_in());
41         stage.threshold_in = _ASSIGN_COMB(threshold_in());
42         stage.ready_in = _ASSIGN(source_out.ready_out());
43         source_out.valid_in = _ASSIGN(stage.valid_out());
44         source_out.data_in = _ASSIGN_COMB(word_comb_func());
45         stage._assign();
46     }
47
48     void _work(bool reset)
49     {
50         stage._work(reset);
51     }
52
53     void _strobe()
54     {
55         stage._strobe();
56     }
57 };
58
59 class QueueEndpoint : public Module
60 {
61     MemoryQueue<8> queue;
62 public:
63     StreamIf<16> sink_in;
64     StreamIf<16> source_out;
65
66     void _assign()
67     {
68         queue.valid_in = _ASSIGN(sink_in.valid_in());
69         queue.data_in = _ASSIGN_COMB(sink_in.data_in());
70         queue.ready_in = _ASSIGN(source_out.ready_out());
71         sink_in.ready_out = _ASSIGN(queue.ready_out());
72         source_out.valid_in = _ASSIGN(queue.valid_out());
73         source_out.data_in = _ASSIGN_COMB(queue.data_out());
74         queue._assign();
75     }
76
77     void _work(bool reset)
78     {
79         queue._work(reset);

```

```

80     }
81
82     void _strobe()
83     {
84         queue._strobe();
85     }
86 };
87
88 class CountedQueue : public QueueEndpoint
89 {
90 public:
91     _PORT(u<32>) sent_out = _ASSIGN_REG(sent_reg);
92
93 private:
94     reg<u<32>> sent_reg;
95
96 public:
97     void _work(bool reset)
98     {
99         QueueEndpoint::_work(reset);
100         if (reset) {
101             sent_reg._next = 0;
102         }
103         else if (source_out.valid_in() && source_out.ready_out()) {
104             sent_reg._next = sent_reg + 1;
105         }
106     }
107
108     void _strobe()
109     {
110         QueueEndpoint::_strobe();
111         sent_reg.strobe();
112     }
113 };
114
115 class StreamMonitor : public Module
116 {
117 public:
118     StreamIf<16> sink_in;
119     _PORT(bool) ready_in;
120     _PORT(bool) valid_out = _ASSIGN(sink_in.valid_in());
121     _PORT(logic<16>) data_out = _ASSIGN_COMB(sink_in.data_in());
122
123     void _assign()
124     {
125         sink_in.ready_out = _ASSIGN(ready_in());
126     }
127
128     void _work(bool) {}
129 };
130
131 class InterfaceTelemetry : public Module
132 {
133     SampleSource producer;

```

```

134     CountedQueue buffer;
135     StreamMonitor monitor;
136
137 public:
138     _PORT(bool) en_in;
139     _PORT(bool) valid_in;
140     _PORT(u<8>) value_in;
141     _PORT(bool) ready_in;
142     _PORT(bool) ready_out = _ASSIGN(producer.ready_out());
143     _PORT(bool) valid_out = _ASSIGN(monitor.valid_out());
144     _PORT(logic<16>) data_out = _ASSIGN_COMB(monitor.data_out());
145     _PORT(u<32>) sent_out = _ASSIGN_COMB(buffer.sent_out());
146
147     void _assign()
148     {
149         producer.en_in = _ASSIGN(en_in());
150         producer.valid_in = _ASSIGN(valid_in());
151         producer.value_in = _ASSIGN_COMB(value_in());
152         producer.threshold_in = _ASSIGN(u<8>(128));
153         monitor.ready_in = _ASSIGN(ready_in());
154         assignIf(producer, buffer, producer.source_out, buffer.sink_in);
155         assignIf(buffer, monitor, buffer.source_out, monitor.sink_in);
156     }
157
158     void _work(bool reset)
159     {
160         producer._work(reset);
161         buffer._work(reset);
162         monitor._work(reset);
163     }
164
165     void _strobe()
166     {
167         producer._strobe();
168         buffer._strobe();
169         monitor._strobe();
170     }
171 };

```

4.5 How the parent connects its three child modules

InterfaceTelemetry contains three child modules: `producer`, `buffer`, and `monitor`. Its `_assign()` connects them with two `assignIf` calls:

- `producer.source_out` to `buffer.sink_in`.
- `buffer.source_out` to `monitor.sink_in`.

In this listing, endpoint `_assign()` methods define their driven outputs; the parent's two `assignIf` calls connect the interfaces. The direct binding to `queue.valid_in` is a scalar child-port connection, not an interface-field connection. Both are covered by Rule 1.

The counter tests the same `valid && ready` condition as the queue's read operation, so it increments once for each word the consumer accepts.

`StreamMonitor` uses an empty `_work(bool)` and inherits the empty default strobe. This adapter adds no latency to chapter 3's design.

This example tests `CountedQueue` inheriting interfaces and a child module. It does not test rebinding inherited scalar ports or accessing template parameters through several base classes.

Forwarding through a module boundary

To forward a wrapper's `proxy_in` interface to its child, put this call in the wrapper's `_assign()`. Here `child` is a direct member of the wrapper, and `proxy_in` and `child.sink_in` have the same interface type:

```
1 assignIf(*this, child, proxy_in, child.sink_in);
```

Here both interface names end in `_in`, so their fields have the same directions. The wrapper forwards its own interface to its child. This differs from connecting a producer's output interface to a consumer's input interface in sibling modules. See `AssignIfHierarchyProxy.cpp` for a complete example.

4.6 Test the interface-based buffer with the existing testbench

We changed the connections and added a counter; sample delivery should still behave as in chapter 3. Reuse `buffer_test.cpp` to check this without rewriting its stimulus or scoreboard. Compile it with `-DUSE_INTERFACES` so `Design` refers to `InterfaceTelemetry` instead of `TelemetryBuffer`. Then convert the new design and lint its generated SystemVerilog.

```
1 g++ -std=c++17 -O2 -DUSE_INTERFACES -I"$CPPHDL_SRC/include" \  
2   "$BOOK/buffer_test.cpp" -o "$BOOK/interface_test" \  
3   "$BOOK/interface_test" \  
4   \  
5   "$CPPHDL" "$BOOK/InterfaceTelemetry.h" \  
6     --generated-dir="$BOOK/sv_interfaces" -- \  
7     -I"$CPPHDL_SRC/include" -I"$BOOK" \  
8   \  
9   verilator --lint-only --top-module InterfaceTelemetry \  
10     "$BOOK/sv_interfaces/Predef_pkg.sv" \  
11     "$BOOK/sv_interfaces/SampleWord_pkg.sv" \  
12     "$BOOK/sv_interfaces/SampleStage.sv" \  
13     "$BOOK/sv_interfaces/MemoryQueue.sv" \  
14     "$BOOK/sv_interfaces/SampleSource.sv" \  
15     "$BOOK/sv_interfaces/QueueEndpoint.sv" \
```

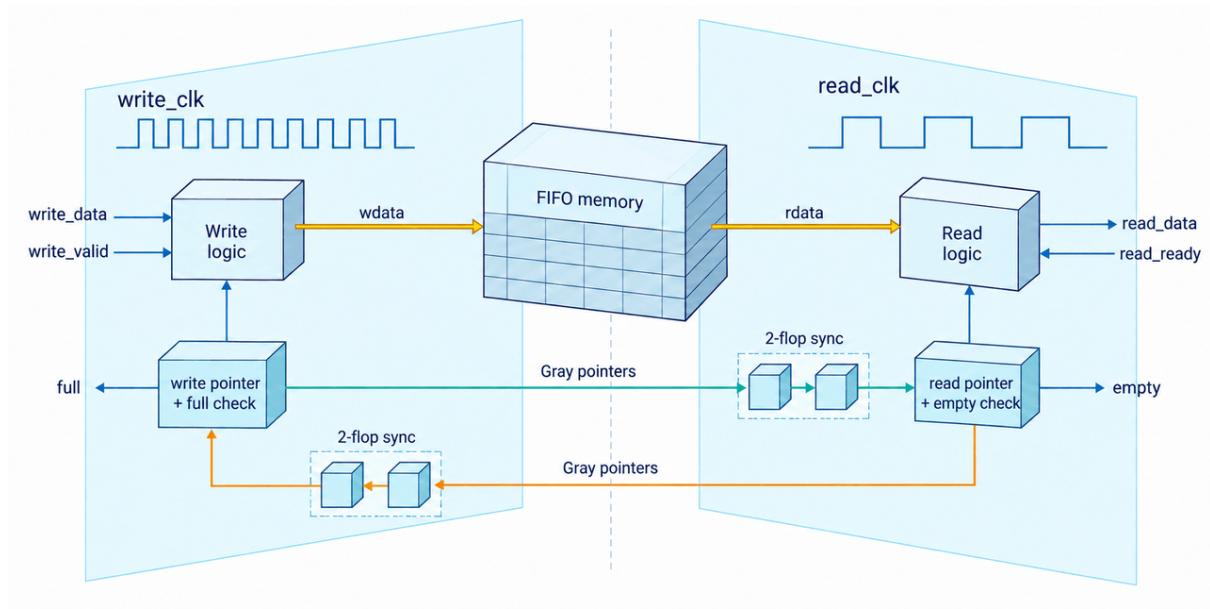
```
16 "$BOOK/sv_interfaces/CountedQueue.sv" \  
17 "$BOOK/sv_interfaces/StreamMonitor.sv" \  
18 "$BOOK/sv_interfaces/InterfaceTelemetry.sv"
```

This build also checks the added counter's final value, `sent_out == 128`, before printing `PASS: 128 ordered samples under backpressure`. The Verilator command is still lint-only; chapter 5 adds RTL execution.

For a type-parameter example, see `TemplateHelperInstantiation.cpp`: an RTL module instantiates a helper as `TemplateInstantiationHelper<TYPE>` and calls its method, which uses `TYPE::BIAS`. This is the type-dependent C++ generation discussed in the introduction, beyond selecting a numeric width.

Checkpoint for the reader: Which class drives `buffer.source_out.ready_out`? Why does `CountedQueue` call both the base work and base strobe methods? Which generated instance corresponds to the inherited queue member?

Next, we add CppHDL's named clock methods and testbench scheduling for CDC.



5. Cross Clock Domains and Test Both Flows

The FIFO uses a conventional Gray-pointer CDC architecture. The new material is how CppHDL assigns methods to clocks, how a native test schedules coincident edges, and how one C++ test can exercise both native and generated RTL models. The FIFO is tested separately from chapter 4's adapters.

5.1 Separate the architecture from its C++ execution

Use `write_clk` for writes and `read_clk` for reads. Replace chapter 3's shared occupancy count with local pointers and synchronized remote pointers; retain 16-bit memory words. The two-clock adapter applies Rule 3 to coincident edges.

5.2 Scope of this example

The example implements a memory-backed FIFO with two-stage pointer synchronizers. CppHDL examples for single-bit, toggle, and mailbox crossings are in `TwoClocksCdc.cpp`. CppHDL does not choose a CDC architecture for you or simulate analog metastability.

5.3 Give every register one clock owner

For 16 rows, use four address bits and five-bit binary/Gray pointers. The diagram shows which clock updates each pointer and synchronizer:

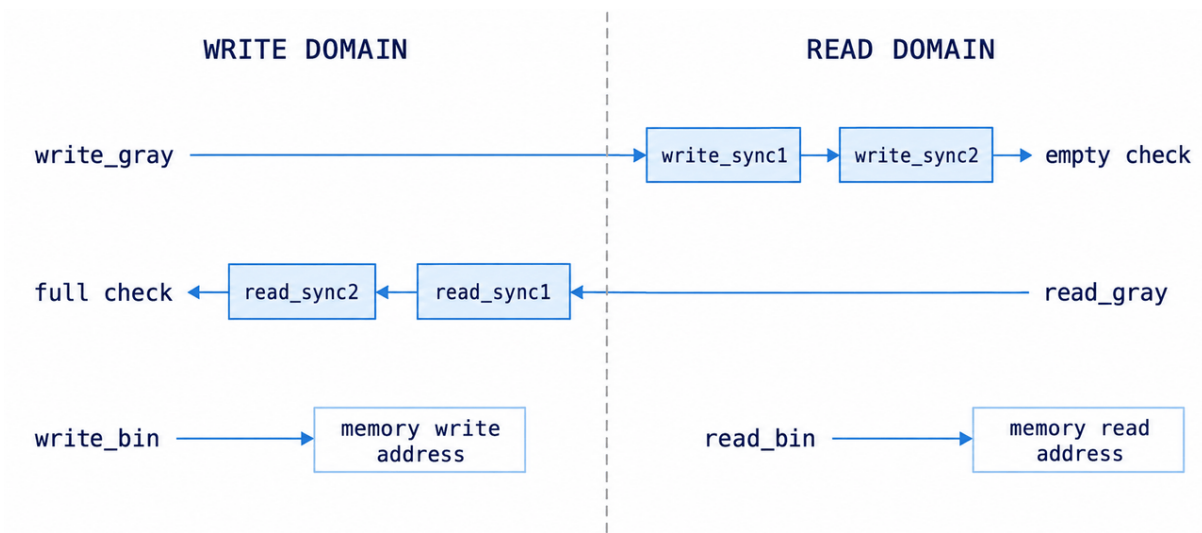


Figure 3: Gray pointers cross into the opposite clock domain through two synchronizer stages; binary pointers address local memory ports.

Each side uses its own clock to update the synchronizers shown next to its check. In the code, `read_sync1/2` are **write-clocked** registers sampling the read pointer; `write_sync1/2` are **read-clocked** registers sampling the write pointer. Their names identify the information being synchronized, not the clock that drives them.

Use this table to review both the C++ methods and generated edge blocks:

Clock	What it updates
Write clock	Write binary/Gray pointers, read-pointer synchronizer stages, write release pipeline, memory write port
Read clock	Read binary/Gray pointers, write-pointer synchronizer stages, read release pipeline
Neither	Ready, valid, and the oldest word are calculated combinationally; they are not separate registers

5.4 Map the pointer equations to the listing

Each clock domain sends a Gray-coded pointer to the other domain. Calculate that pointer from the local binary pointer with:

```
1 gray = binary XOR (binary >> 1)
```

The empty test compares `read_gray_reg` with `write_sync2_reg`. The full test compares `write_gray_reg` with `read_sync2_reg` after inverting its top two bits. The full mask is

0b11000 for this depth and extra-wrap-bit scheme. These comparisons use current synchronizer values, as specified by Rule 2.

5.5 Decide how both sides reset and restart

During reset, the test calls `_work_write_clk(true)` on write-clock edges and `_work_read_clk(true)` on read-clock edges. Both sides reset their pointers, which discards the queued data.

Reset takes effect on each side's clock edge. Keep it high long enough for both clocks to sample it; the test counts no transfers during reset. After reset goes low, each side uses a two-bit register to delay transfers by two local edges:

Local edge after reset goes low	Value read by work	Value after strobe	Result
First	00	01	Local pointers remain reset
Second	01	11	Pointers stay at zero; bit 1 now permits transfers on later edges
Third	11	11	A handshake can advance the local pointer

Work reads the current register value, so it allows transfers starting with the third edge, subject to ready and valid. This example does not support resetting only one side. If a clock stops, that side cannot reset until the clock restarts. Section 5.7 shows handlers for asynchronous reset instead.

5.6 Implement the two-clock FIFO

We can now combine the pointer checks, memory access, and reset logic into `AsyncSamples`. Its write method accepts words on write-clock edges; its read method removes words on read-clock edges. The test uses these handshake conditions to decide which transfers to check:

Local edge	Acceptance expression outside reset
Write rising edge	<code>write_valid_in() && write_ready_out()</code>
Read rising edge	<code>read_valid_out() && read_ready_in()</code>

The method suffix selects the clock: `_work_write_clk` and `_strobe_write_clk` form the write-clock pair; the `_read_clk` pair handles the read side. Their responsibilities

are unchanged from Rules 2 and 3. Section 5.7 shows how to declare these clocks to the converter. The write method tests `write_release_reg[1]`; the read method tests `read_release_reg[1]`. Each keeps its pointers at zero while that bit is zero.

CppHDL passes the `ASYNC_REG` comments into generated RTL attributes, including those on the release pipelines. They do not change native simulation behavior.

`read_data_comb_func()` reads memory only when `read_valid_comb_func()` returns true; otherwise it returns zero. These are combinational reads, not synchronous block-RAM reads.

File: AsyncSamples.h

```

1  #pragma once
2  #include <cpphdl.h>
3  using namespace cpphdl;
4
5  class AsyncSamples : public Module
6  {
7  public:
8      _PORT(bool) write_valid_in;
9      _PORT(logic<16>) write_data_in;
10     _PORT(bool) write_ready_out = _ASSIGN_COMB(write_ready_comb_func());
11     _PORT(bool) read_ready_in;
12     _PORT(bool) read_valid_out = _ASSIGN_COMB(read_valid_comb_func());
13     _PORT(logic<16>) read_data_out = _ASSIGN_COMB(read_data_comb_func());
14
15 private:
16     static constexpr size_t DEPTH = 16;
17     static constexpr size_t ADDR_BITS = 4;
18     static constexpr size_t PTR_BITS = ADDR_BITS + 1;
19     memory<u8, 2, DEPTH> storage;
20
21     reg<u<PTR_BITS>> write_bin_reg;
22     reg<u<PTR_BITS>> write_gray_reg;
23     // (* ASYNC_REG = "TRUE" *)
24     reg<u<PTR_BITS>> read_sync1_reg;
25     // (* ASYNC_REG = "TRUE" *)
26     reg<u<PTR_BITS>> read_sync2_reg;
27     // (* ASYNC_REG = "TRUE" *)
28     reg<u<2>> write_release_reg;
29
30     reg<u<PTR_BITS>> read_bin_reg;
31     reg<u<PTR_BITS>> read_gray_reg;
32     // (* ASYNC_REG = "TRUE" *)
33     reg<u<PTR_BITS>> write_sync1_reg;
34     // (* ASYNC_REG = "TRUE" *)
35     reg<u<PTR_BITS>> write_sync2_reg;
36     // (* ASYNC_REG = "TRUE" *)
37     reg<u<2>> read_release_reg;
38
39     bool write_ready_comb;
40     bool& write_ready_comb_func()
41     {

```

```

42     write_ready_comb = ((uint32_t)write_release_reg & 2u) != 0
43         && write_gray_reg !=
44             (read_sync2_reg ^ u<PTR_BITS>((1u << ADDR_BITS)
45                 | (1u << (ADDR_BITS - 1))));
46     return write_ready_comb;
47 }
48
49 bool read_valid_comb;
50 bool& read_valid_comb_func()
51 {
52     read_valid_comb = ((uint32_t)read_release_reg & 2u) != 0
53         && read_gray_reg != write_sync2_reg;
54     return read_valid_comb;
55 }
56
57 logic<16> read_data_comb;
58 logic<16>& read_data_comb_func()
59 {
60     read_data_comb = 0;
61     if (read_valid_comb_func()) {
62         read_data_comb = (logic<16>)storage[
63             (uint32_t)read_bin_reg & (uint32_t)(DEPTH - 1)];
64     }
65     return read_data_comb;
66 }
67
68 public:
69     void _assign() {}
70
71     void _work_write_clk(bool reset)
72     {
73         u<PTR_BITS> next;
74         if (reset) {
75             write_release_reg._next = 0;
76         }
77         else {
78             write_release_reg._next = (write_release_reg << 1) | 1;
79         }
80         if (reset || ((uint32_t)write_release_reg & 2u) == 0) {
81             write_bin_reg._next = 0;
82             write_gray_reg._next = 0;
83             read_sync1_reg._next = 0;
84             read_sync2_reg._next = 0;
85         }
86         else {
87             read_sync1_reg._next = read_gray_reg;
88             read_sync2_reg._next = read_sync1_reg;
89             if (write_valid_in() && write_ready_comb_func()) {
90                 storage[(uint32_t)write_bin_reg & (uint32_t)(DEPTH - 1)]
91                     = write_data_in();
92                 next = write_bin_reg + 1;
93                 write_bin_reg._next = next;
94                 write_gray_reg._next = next ^ (next >> 1);
95             }

```

```

96     }
97 }
98
99 void _strobe_write_clk()
100 {
101     storage.apply();
102     write_release_reg.strobe();
103     write_bin_reg.strobe();
104     write_gray_reg.strobe();
105     read_sync1_reg.strobe();
106     read_sync2_reg.strobe();
107 }
108
109 void _work_read_clk(bool reset)
110 {
111     u<PTR_BITS> next;
112     if (reset) {
113         read_release_reg._next = 0;
114     }
115     else {
116         read_release_reg._next = (read_release_reg << 1) | 1;
117     }
118     if (reset || ((uint32_t)read_release_reg & 2u) == 0) {
119         read_bin_reg._next = 0;
120         read_gray_reg._next = 0;
121         write_sync1_reg._next = 0;
122         write_sync2_reg._next = 0;
123     }
124     else {
125         write_sync1_reg._next = write_gray_reg;
126         write_sync2_reg._next = write_sync1_reg;
127         if (read_ready_in() && read_valid_comb_func()) {
128             next = read_bin_reg + 1;
129             read_bin_reg._next = next;
130             read_gray_reg._next = next ^ (next >> 1);
131         }
132     }
133 }
134
135 void _strobe_read_clk()
136 {
137     read_release_reg.strobe();
138     read_bin_reg.strobe();
139     read_gray_reg.strobe();
140     write_sync1_reg.strobe();
141     write_sync2_reg.strobe();
142 }
143 };

```

5.7 Relate the methods to generated RTL

Generate SystemVerilog with separate write-clock and read-clock processes. Pass both clock names and frequencies to the converter so it can match them to the methods in `AsyncSamples`:

```
1 "$CPPHDL" "$BOOK/AsyncSamples.h" \  
2   --generated-dir="$BOOK/sv_async" \  
3   --primary_clock write_clk 100000000 \  
4   --secondary_clock read_clk 71428571 -- \  
5   -I"$CPPHDL_SRC/include"
```

Declare the fastest clock as primary. Declaring any secondary clock also requires an explicit primary declaration. Frequencies are in Hz; the read frequency here corresponds approximately to a 14 ns period. The testbench still determines the exact edge times and reset duration.

For a multi-clock design the positive-edge pair `_work_<clock>(bool reset)` and `_strobe_<clock>()` is required for each declared clock. Optional negative-edge pairs use `_work_neg_<clock>` and `_strobe_neg_<clock>` for registers updated on falling edges.

Inspect `AsyncSamples.sv`. It should have clock ports and separate blocks with these sensitivities:

```
1 always_ff @(posedge write_clk)  
2 always_ff @(posedge read_clk)
```

These fragments show which edge triggers each block; they are not complete blocks. Each generated block calls the work task for that clock and applies the resulting register updates. Only the write-clock process writes to `storage`.

Integrating the earlier modules

Clock names currently apply to the whole design; the converter does not map different child clock names to parent clock names. To add the earlier stage to this design, give it and its descendants methods for both `write_clk` and `read_clk`.

For a stage that uses only `write_clk`, put its existing calculations and register updates in the write-clock methods and leave its read-clock methods empty. Do the reverse for a component that uses only `read_clk`. The parent's `_work_write_clk()` calls the children's `_work_write_clk()` methods, and so on for the other three methods, in Rule 3's order.

Use `assignIf` for connections within each clock domain, not to replace the FIFO between domains. It connects signals but adds no synchronizers.

When true asynchronous assertion is required

The current FIFO resets only on clock edges. To reset it even when a clock is stopped, add an asynchronous reset handler for each domain. The write-side handler assigns the reset values without waiting for a write-clock edge:

```
1 void _reset_pos_write_clk()
2 {
3     write_bin_reg._next = 0;
4     write_gray_reg._next = 0;
5     read_sync1_reg._next = 0;
6     read_sync2_reg._next = 0;
7     write_release_reg._next = 0;
8 }
```

This is an optional modification, **not part of the tested FIFO listing**. A corresponding read-domain handler must reset every register updated by the read clock. The converter adds `posedge reset` to the clocked block's sensitivity list, so the block also runs when reset rises. It then calls either the reset handler or the normal work task. The `pos` in the method name describes the clock edge, not the reset's polarity; the reset signal is active-high.

For asynchronous reset, the C++ test uses these calls:

Event	What the C++ test calls
Reset rises, with or without a clock edge	Call all affected reset handlers, then their strobes
Clock edge while reset remains high	Call that clock's reset handler instead of work, then strobe
Reset falls	No work or strobe call from deassertion alone

If reset rises for both sides, call both reset handlers, then both strobe methods, following Rule 3's call order.

The fragment belongs inside `AsyncSamples` and requires the adapter changes in this table. The complete listings and commands continue to test clocked reset, not this alternative.

Keep the two-stage release delay on each side and reset both sides together. Making reset asynchronous does not make it safe to reset only one side. See the asynchronous reset tests and the specification for the complete API behavior.

5.8 Run the same transfer test against C++ and SystemVerilog

Native execution tests the C++ model. Verilator execution tests the generated SystemVerilog. Running both is valuable because a working native model alone does not establish that conversion preserved widths, direction, reset, or scheduling.

The `TestModel` class below provides the same methods for either implementation:

- For the native model, it changes the C++ variables bound to input ports and reads outputs through port getters.
- For the Verilated model, inputs and outputs are generated class members; `eval()` evaluates the RTL after inputs change.
- The stimulus, expected-data queue, and assertions are shared.

Defining `VERILATOR` selects the generated `VAsyncSamples` class; otherwise, the adapter instantiates `AsyncSamples` directly. The host scoreboard needs no simulator-specific code.

The test uses `t % 10 == 0` for write-clock rising edges and `t % 14 == 0` for read-clock rising edges. Both conditions are true every 70 ns.

`wr` and `rd` select which work methods to call. `wlevel` and `rlevel` hold the clock levels assigned to the Verilated model. The loop increments `t` by one nanosecond per iteration. It lowers the Verilator clock inputs between rising edges; the native model has no falling-edge methods in this example.

`TestModel` implements Rule 3 for both models:

Step	Native CppHDL	Verilated RTL
Drive inputs	Update bound variables and advance <code>_system_clock</code>	Assign input members and call <code>eval()</code>
Observe transfers	Read port getters before work	Read output members before changing clocks
Apply edges	Call selected work methods, then selected strobos; advance <code>_system_clock</code>	Assign both clock levels, then call <code>eval()</code> once

For example, when `wr` and `rd` are both true, `edges()` calls write work, read work, write strobe, then read strobe.

Reset is active for `0 <= t < 35` and `1800 <= t < 1835`, covering both clocks. At the second reset, the test checks that words remain queued, then clears `expected`, the transfer counts, `saw_full`, and `saw_empty`. Afterward, the test requires 512 words in order and checks that the FIFO became both full and empty. It runs these assertions against each model.

When both handshakes occur at one timestamp, the scoreboard checks and removes its oldest expected word before adding the newly written word. Otherwise, that new word might incorrectly satisfy a read when the FIFO had actually been empty.

File: `async_test.cpp`

```

1  #include <cassert>
2  #include <cstdint>
3  #include <cstdio>
4  #include <deque>
5
6  #ifdef VERILATOR
7  #include "VAsyncSamples.h"
8  #else
9  #include "AsyncSamples.h"
10 long _system_clock = -1;
11 #endif
12
13 class TestModel
14 {
15 #ifdef VERILATOR
16     VAsyncSamples dut;
17 #else
18     AsyncSamples dut;
19     bool write_valid = false;
20     cpphdl::logic<16> write_data = 0;
21     bool read_ready = false;
22 #endif
23
24 public:
25     TestModel()
26     {
27 #ifdef VERILATOR
28         dut.write_clk = 0;
29         dut.read_clk = 0;
30 #else
31         dut.write_valid_in = _ASSIGN(write_valid);
32         dut.write_data_in = _ASSIGN_REG(write_data);
33         dut.read_ready_in = _ASSIGN(read_ready);
34         dut._assign();
35 #endif
36     }
37
38     void drive(bool reset, bool valid, uint16_t data, bool ready)
39     {
40 #ifdef VERILATOR
41         dut.reset = reset;
42         dut.write_valid_in = valid;
43         dut.write_data_in = data;
44         dut.read_ready_in = ready;
45         dut.eval();
46 #else
47         (void)reset;
48         write_valid = valid;
49         write_data = data;
50         read_ready = ready;
51         ++_system_clock;
52 #endif
53     }

```

```

54
55     bool can_write()
56     {
57 #ifdef VERILATOR
58         return dut.write_ready_out;
59 #else
60         return dut.write_ready_out();
61 #endif
62     }
63
64     bool can_read()
65     {
66 #ifdef VERILATOR
67         return dut.read_valid_out;
68 #else
69         return dut.read_valid_out();
70 #endif
71     }
72
73     uint16_t data()
74     {
75 #ifdef VERILATOR
76         return dut.read_data_out;
77 #else
78         return (uint16_t)dut.read_data_out();
79 #endif
80     }
81
82     void edges(bool reset, bool wr, bool rd, bool wlevel, bool rlevel)
83     {
84 #ifdef VERILATOR
85         (void)wr;
86         (void)rd;
87         dut.write_clk = wlevel;
88         dut.read_clk = rlevel;
89         dut.eval();
90 #else
91         (void)wlevel;
92         (void)rlevel;
93         if (wr) dut._work_write_clk(reset);
94         if (rd) dut._work_read_clk(reset);
95         if (wr) dut._strobe_write_clk();
96         if (rd) dut._strobe_read_clk();
97         ++_system_clock;
98 #endif
99     }
100 };
101
102 static uint16_t pattern(unsigned n)
103 {
104     return (uint16_t)((n * 73u) ^ 0xa55au);
105 }
106
107 int main()

```

```

108 {
109     TestModel model;
110     std::deque<uint16_t> expected;
111     unsigned sent = 0;
112     unsigned received = 0;
113     bool saw_full = false;
114     bool saw_empty = false;
115     bool wlevel = false;
116     bool rlevel = false;
117
118     for (unsigned t = 0; t < 30000; ++t) {
119         const bool reset = t < 35 || (t >= 1800 && t < 1835);
120         const bool wr = t % 10 == 0;
121         const bool rd = t % 14 == 0;
122         if (t == 1800) {
123             assert(!expected.empty());
124             expected.clear();
125             sent = received = 0;
126             saw_full = saw_empty = false;
127         }
128         const bool valid = !reset && sent < 512;
129         const bool ready = !reset && t >= 500 && t % 98 >= 28;
130         model.drive(reset, valid, pattern(sent), ready);
131
132         // Observe the old state before either domain commits this timestamp.
133         if (!reset) {
134             const bool push = wr && valid && model.can_write();
135             const bool pop = rd && ready && model.can_read();
136             if (wr && valid && !model.can_write() && expected.size() == 16) {
137                 saw_full = true;
138             }
139             if (rd && !model.can_read()) saw_empty = true;
140             if (pop) {
141                 assert(!expected.empty());
142                 assert(model.data() == expected.front());
143                 expected.pop_front();
144                 ++received;
145             }
146             if (push) {
147                 expected.push_back(pattern(sent));
148                 ++sent;
149             }
150         }
151
152         if (wr) wlevel = true;
153         else if (t % 10 == 5) wlevel = false;
154         if (rd) rlevel = true;
155         else if (t % 14 == 7) rlevel = false;
156         model.edges(reset, wr, rd, wlevel, rlevel);
157
158         if (t > 1835 && sent == 512 && received == 512) {
159             assert(expected.empty() && saw_full && saw_empty);
160             std::puts("PASS: 512 ordered words after coordinated reset");
161             return 0;

```

```

162     }
163 }
164 std::fputs("FAIL: CDC queue did not drain\n", stderr);
165 return 1;
166 }

```

5.9 Build and run both flows

First compile and execute the native model:

```

1 g++ -std=c++17 -O2 -I"$CPPHDL_SRC/include" \
2     "$BOOK/async_test.cpp" -o "$BOOK/async_native"
3 "$BOOK/async_native"

```

Then generate RTL with the command in section 5.7 and build the Verilated model:

```

1 verilogator --cc --exe --build -j 2 \
2     --top-module AsyncSamples --Mdir "$BOOK/obj_async" \
3     -Wno-fatal \
4     "$BOOK/sv_async/Predef_pkg.sv" \
5     "$BOOK/sv_async/AsyncSamples.sv" \
6     "$BOOK/async_test.cpp" \
7     -CFLAGS "-std=c++17 -DVERILATOR" \
8     -MAKEFLAGS "CXX=g++ LINK=g++"
9 "$BOOK/obj_async/VAsyncSamples"

```

Use an absolute path to the testbench because Make runs the generated build rules from the object directory. `CXX=g++ LINK=g++` selects the compiler and linker driver. Omit this override if Verilator is already configured with the correct compiler, or substitute your supported compiler.

Both successful runs print:

```

1 PASS: 512 ordered words after coordinated reset

```

These commands run both models, unlike the lint-only RTL commands in earlier chapters. For automation, register separate native and RTL CTest entries and regenerate the RTL and rebuild its executable after changes. Passing both tests shows that both models satisfy these assertions for these inputs; it does not prove equivalence for all inputs.

Read warnings instead of hiding them

During validation with Verilator 5.050, this CDC example produced MULTIDRIVEN diagnostics on converter-generated `*_tmp` next-state variables assigned in both a work task and the clocked block that calls it. The C++ and RTL tests passed with `-Wno-fatal`, which allows simulation despite warnings. That is **not a warning-free lint result or approval of the physical CDC implementation**.

The command keeps diagnostics visible rather than disabling their warning class. For each reported assignment, check which clock is supposed to update that register, using the table in section 5.3. If two domains really do write one register, fix the design. The warnings about these generated temporary variables are not a reason to ignore other multiple-driver warnings.

5.10 What this test checks, and what to add next

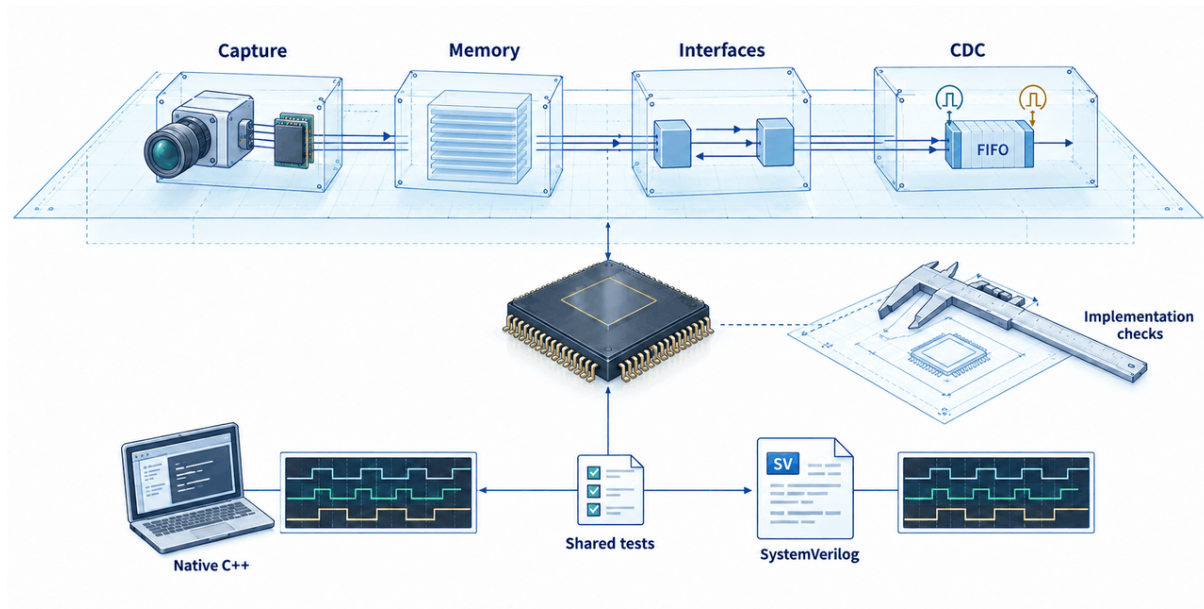
Before reusing this FIFO, distinguish what the supplied test checks from what still needs testing. The table links each covered behavior to its stimulus or assertion; the suggestions below cover additional cases.

What we check	How the test checks it
Edges occurring separately and together	Use 10 ns and 14 ns periods; sample both handshakes before applying coincident edges
The writer must wait when full	Write faster than reading; delay and periodically pause reads
Reads after starting empty	Resume after reset and wait for the write pointer to reach the read domain
Words arrive intact and in order	Compare received words with the expected-data queue
Pointers wrap correctly	Transfer hundreds of words through 16 slots
Reset discards pending data	Reset both sides while words remain queued, then check the restarted run
C++ and RTL pass the same checks	Run the same assertions with each model

Before using the FIFO in a product, also test a faster reader, clocks that start at different times, clocks paused around reset, and other supported depths and widths. Let the source pause between words, and vary the length of consumer pauses randomly. Add an assertion that output valid stays asserted and output data stays unchanged while the consumer is blocked. These are useful extensions, not claims about what the listing already tests.

CppHDL adds no exemption from normal CDC signoff: metastability, placement, timing constraints, reset safety, and memory implementation still require the usual physical-design checks.

Check your CppHDL understanding: Which methods must run when two clocks rise together? Which additional testbench calls would asynchronous reset need? Why does passing the native test not validate the converter output?



6. Conclusion

The circuits are conventional RTL. We compiled their C++ descriptions with testbenches, wrote output values to VCD, added a counter through inheritance, and ran the same FIFO test against C++ and generated SystemVerilog.

The benefit is access to C++ types, compilers, analysis tools, and software test environments without maintaining a separate behavioral model. Follow the work/strobe call order, test the generated RTL, and retain the usual synthesis and CDC checks.